

MC-1: Design and Development of AMTTP — A Four-Layer Real-Time Blockchain Compliance System

**Executive Summary:** I designed and built a four-layer real-time compliance system for blockchain transactions. Traditional AML tools act only after transactions are completed, while AMTTP assesses risk, verifies sanctions, and applies contract decisions in less than a second — before any funds are transferred. In its first public month, the live platform processed **920,848 transactions** and attracted **1,220 unique visitors** at amttp.com. Every layer — consumer app, enterprise dashboard, nine-service AI backend, and blockchain contracts — was built by me.

What this proves for Mandatory Criteria: independent technical leadership, not merely participation. The evidence combines a public deployment, third-party infrastructure analytics, live product screens, and a repeatable architecture showing that I personally took an original digital-technology product from design to public release.

Live Public Deployment Evidence

4-MINUTE VIDEO DEMONSTRATION

[https://youtu.be/I0dXSN\\_ScR0](https://youtu.be/I0dXSN_ScR0)

Public demo: amttp.com | Sepolia-backed | 1,220 unique visitors | 38,320 requests | 5 Apr – 5 May 2026

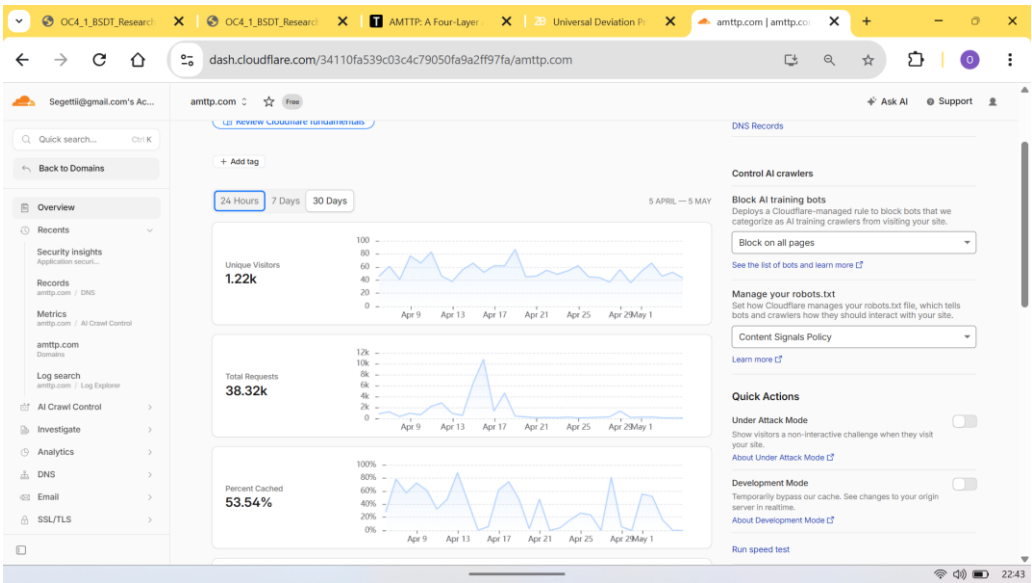


Figure 1: Cloudflare analytics — 1,220 unique visitors, 38,320 requests, 53.54% cache-hit ratio (5 Apr – 5 May 2026). Independently verifiable at amttp.com.

Technical Contributions

- **Pre-settlement enforcement at scale:** 920,848 transactions processed on a live testnet; decisions issued before funds move — not after.
- **Dual-model AI scoring:** gradient-boosted classifier + Graph Neural Network; fraud-detection ROC-AUC 0.9999998.
- **90%+ cost reduction:** compact distilled model runs on standard CPUs, eliminating specialist GPU hardware.
- **Privacy-preserving identity:** sanctions verified on-chain without exposing personal data — GDPR-compliant on a public ledger.
- **Regulatory audit trail:** every compliance decision permanently recorded on-chain (FCA / MiCA requirements).
- **Zero-downtime upgrades:** UUPS upgradeable contract design — compliance rules change without taking the live platform offline.

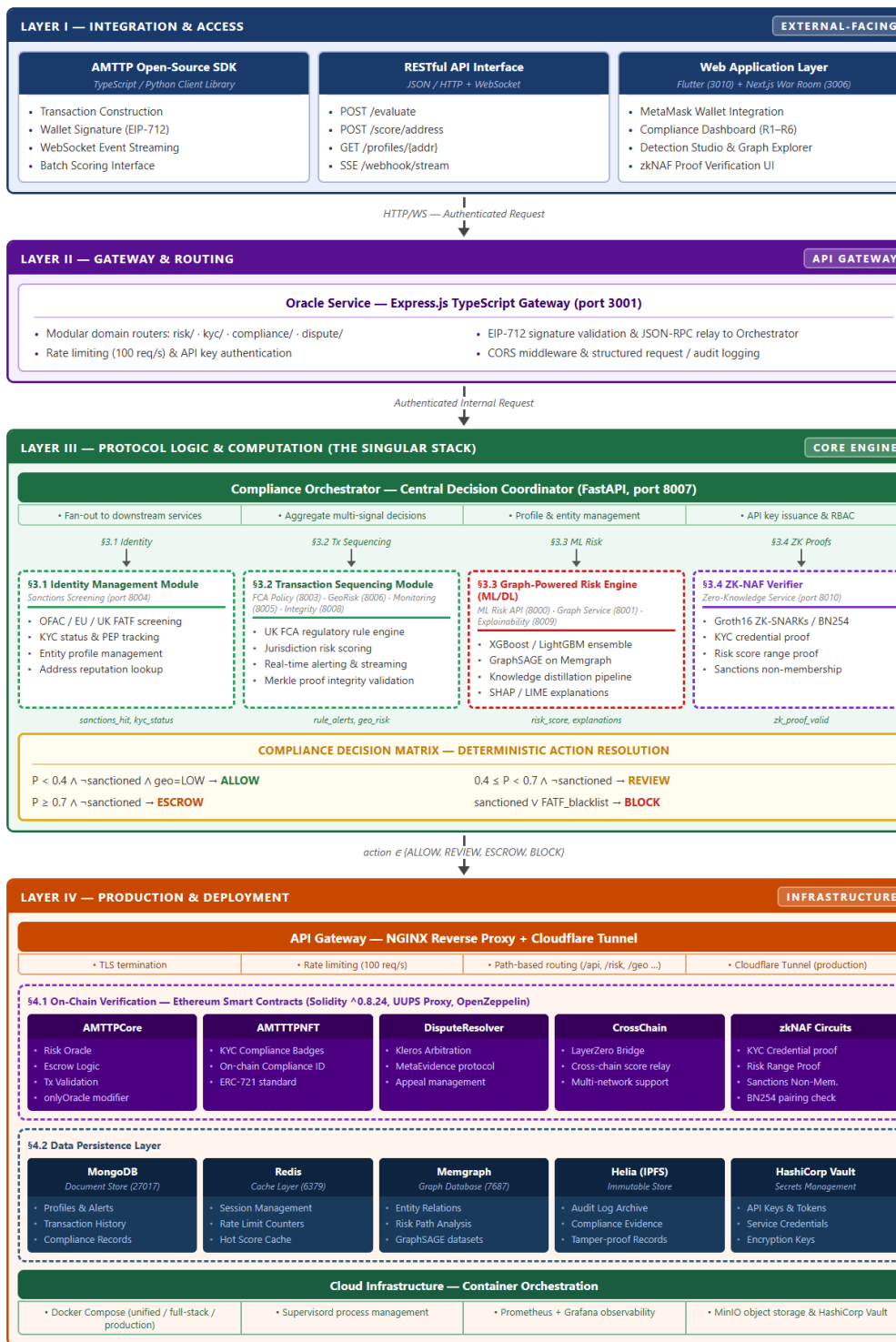


Figure 2: AMTTP four-layer architecture — Flutter + Next.js (UI) → Express Gateway → Python FastAPI orchestrator (9 microservices) → 38 Solidity files, 17 Docker containers. All layers sole-authored.

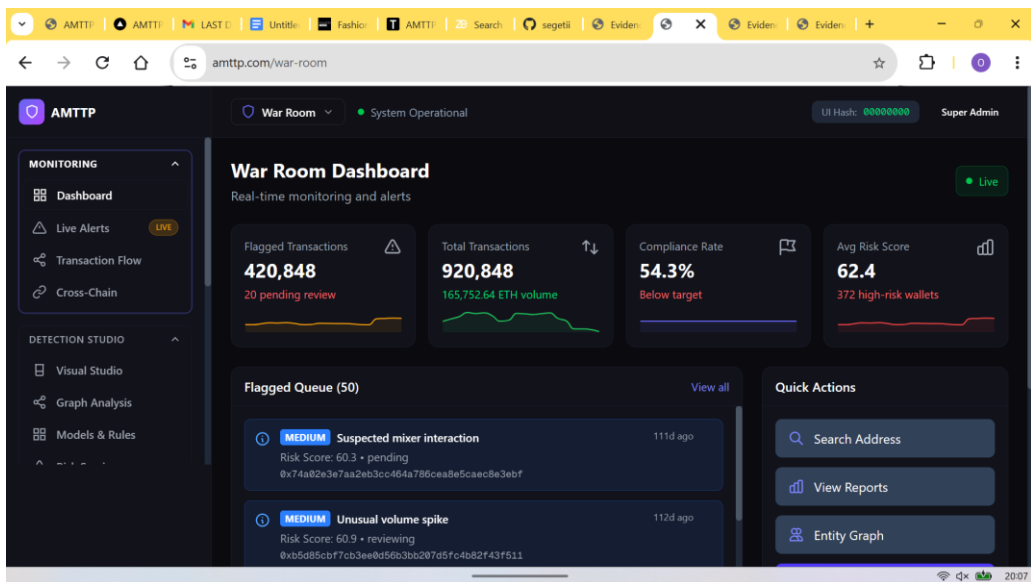


Figure 3: War Room enterprise dashboard (Next.js, port 3006) — live compliance monitoring for compliance officers. Role-based access enforced server-side.

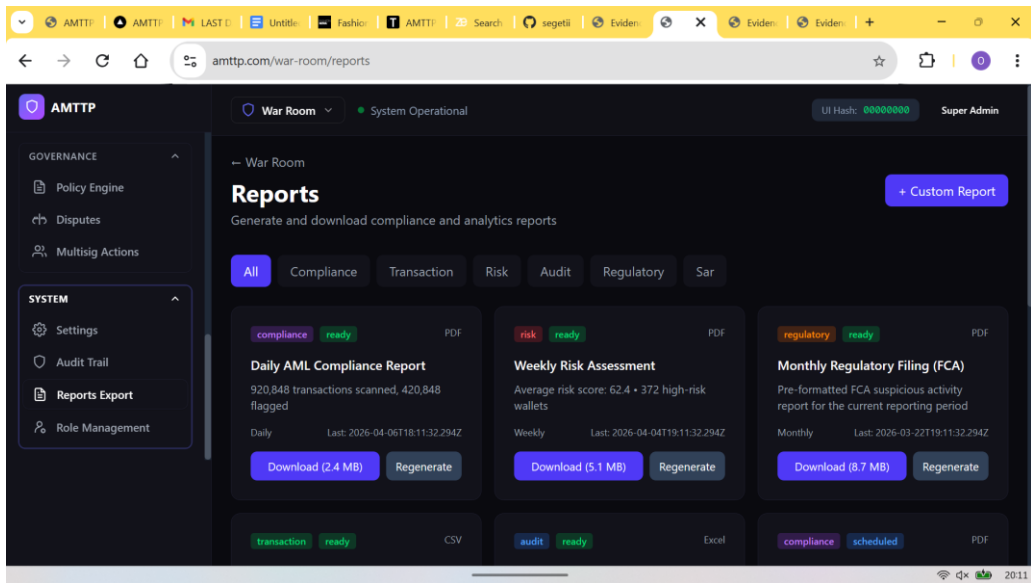


Figure 4: Automated report generation — Daily AML PDF (2.4 MB), Weekly Risk Assessment (5.1 MB), Monthly FCA SAR (8.7 MB).